

Supplementary figures: parameter effects

This appendix is separate from the one-page Task 2 report contribution. It is not automatically part of the five-page group report.

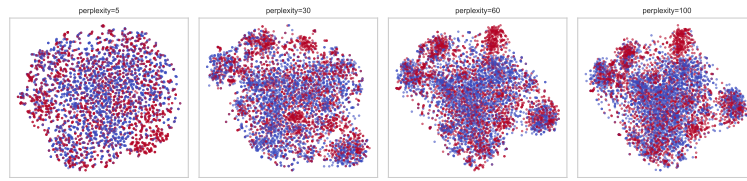


Figure A1. Selected original t-SNE sweep panels with enlarged parameter labels: perplexity 5, 30, 60 and 100. Same fixed 5,000 cells, 29 PCs and seed 42. Blue/cyan denote CMV-negative/positive donor labels. Trustworthiness at 60 is only slightly above the values at 50 and 100.

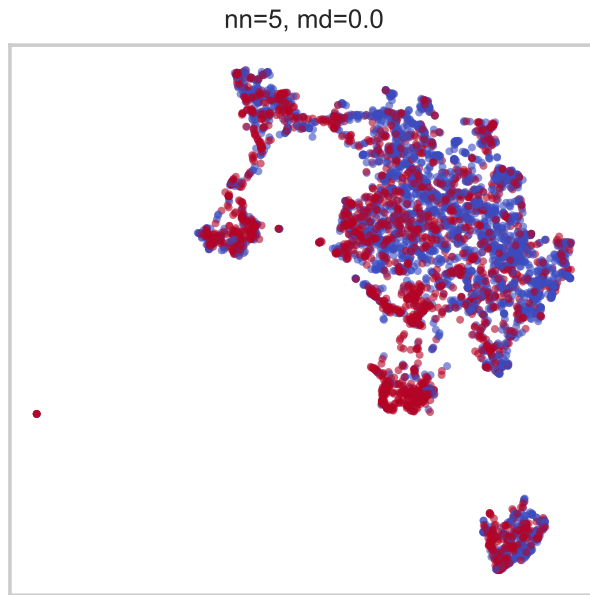


Figure A2. Four settings from the original UMAP grid. Rows contrast 5 with 50 neighbours; columns contrast `min_dist=0` with 0.5. Smaller `min_dist` permits tighter packing; `n_neighbors` changes the graph's neighbourhood scale. Parameter-induced islands are not independently established cell types.

Supplementary figures: marker expression

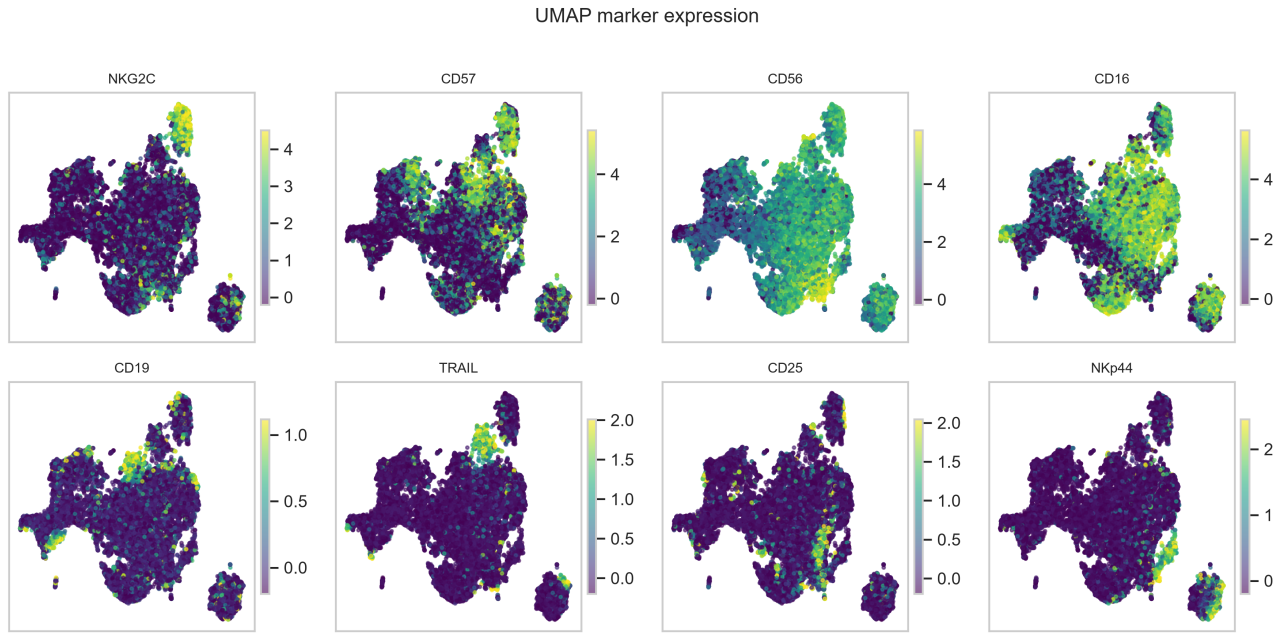


Figure A3. Original UMAP marker overlays from notebook cell 41 (counting from 1), on the balanced 40,000-cell sample. Each panel colours the same points by a different arcsinh-transformed marker; colour limits are clipped at that marker's 1st/99th percentiles. Numerical scales therefore differ between markers.

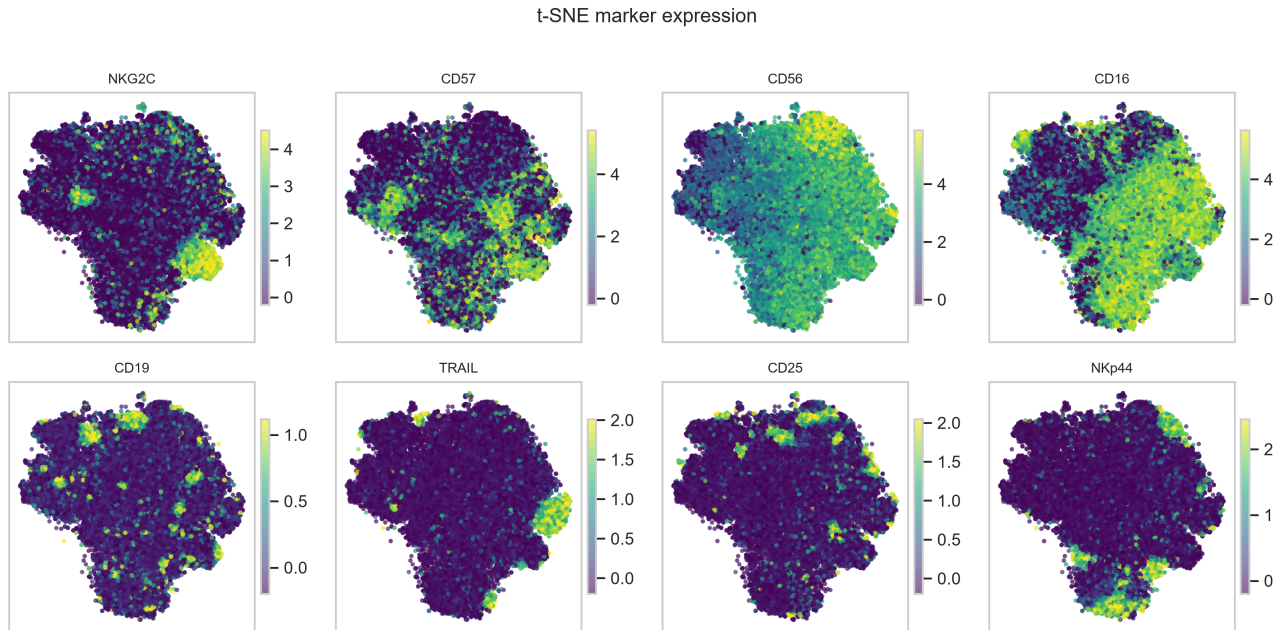


Figure A4. The same markers and cells on the saved t-SNE map. NKG2C/CD57 expression can guide follow-up of a memory-like NK phenotype, but expression gradients alone do not validate a functional adaptive population or label individual cells as CMV-infected. The plots' "full data" titles refer to the full balanced analysis sample, not all 261,593 available NK cells.